

PAPER_925: Quasar Jet Phonon Modulation Factor $M_{\text{jet}}(\Gamma)$

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Abstract

Defines the jet modulation factor $M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} * \exp[-(\Gamma - \Gamma_0)^2 / (2 * \sigma_{\Gamma}^2)]$ with $A_{\text{jet}} = 1.5$, $\Gamma_0 = \omega_{\text{SCm}} = 2\pi 1.25 \text{ THz}$, $\sigma_{\Gamma} = 0.08 \text{ THz}$. The phonon-enhanced Blandford-Znajek jet power is $P_{\text{jet}} = P_{\text{BZ}} * (1 + M_{\text{jet}})$, providing up to 3.5x amplification at resonance. The Γ -sweep reveals a narrow enhancement window (FWHM $\sim 0.19 \text{ THz}$) centered on the SCm phonon frequency, explaining why jet power varies by factors of 2-4 across different AGN despite similar BH masses and spins.

1. Core Equations

Section A: Lagrangian

```
M_jet(Gamma) = 1 + A_jet * exp[-(Gamma - Gamma_0)^2 / (2 * sigma_Gamma^2)]
P_jet = P_BZ * (1 + M_jet)
P_BZ = (pi / (6 * mu_0)) * B^2 * r_g^2 * c * a^2
r_g = G * M / c^2
```

Section B: VDS/DVP/BH Number Systems

```
WSTP export: Mjet[Gamma_] := 1 + 1.5 * Exp[-(Gamma-Gamma0)^2/(2*sigmaGamma^2)]
WSTP export: Pjet[Gamma_] := PBZ * (1 + Mjet[Gamma])
WSTP export: Table[{Gamma, Pjet[Gamma]}, {Gamma, 0.5 THz, 2.0 THz, 0.1 THz}]
```

Section SM: SM Anchors

```
A_jet = 1.5 (jet modulation amplitude)
sigma_Gamma = 0.08 THz (linewidth spread)
Gamma_0 = 1.25 THz (SCm phonon frequency)
FWHM = 2*sqrt(2*ln(2)) * sigma_Gamma ~ 0.19 THz
```

2. Parameters

Parameter	Default	Description
M_{bh}	$6.5e9 M_{\text{sun}}$	BH mass
a_{spin}	0.9	Spin parameter
B_{field}	50 T	Magnetic field
A_{jet}	1.5	Modulation amplitude
$\sigma_{\Gamma} \text{ THz}$	0.08	Linewidth spread
Γ_{THz}	1.25	Phonon linewidth

3. Key Results

Gamma (THz)	M_jet	Enhancement
0.5	1.00	2.00x
1.0	1.38	2.38x
1.25	2.50	3.50x
1.5	1.38	2.38x
2.0	1.00	2.00x

4. Physical Interpretation

The jet modulation factor explains the diversity of AGN jet powers as a function of the local phonon environment. AGN with accretion disk phonon frequencies near 1.25 THz experience maximum jet enhancement (3.5x P_BZ), while those with mismatched frequencies see only the baseline 2x modulation. This provides a testable prediction: AGN jet power should correlate with accretion disk temperature through the phonon frequency relation $\omega \sim k_B * T / \hbar$.

5. References

- PAPER_922: M87 jet power curve (Session 210c)
- PAPER_910: Phonon jet launching M87/Sgr A* (Session 210)
- quasar_jet_phonon.py: 4-class standalone module
- WSTP expressions #29-30